Team SPIKES

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Seeking Predictable Indicators & Knowledge Extracted from CGM Spikes

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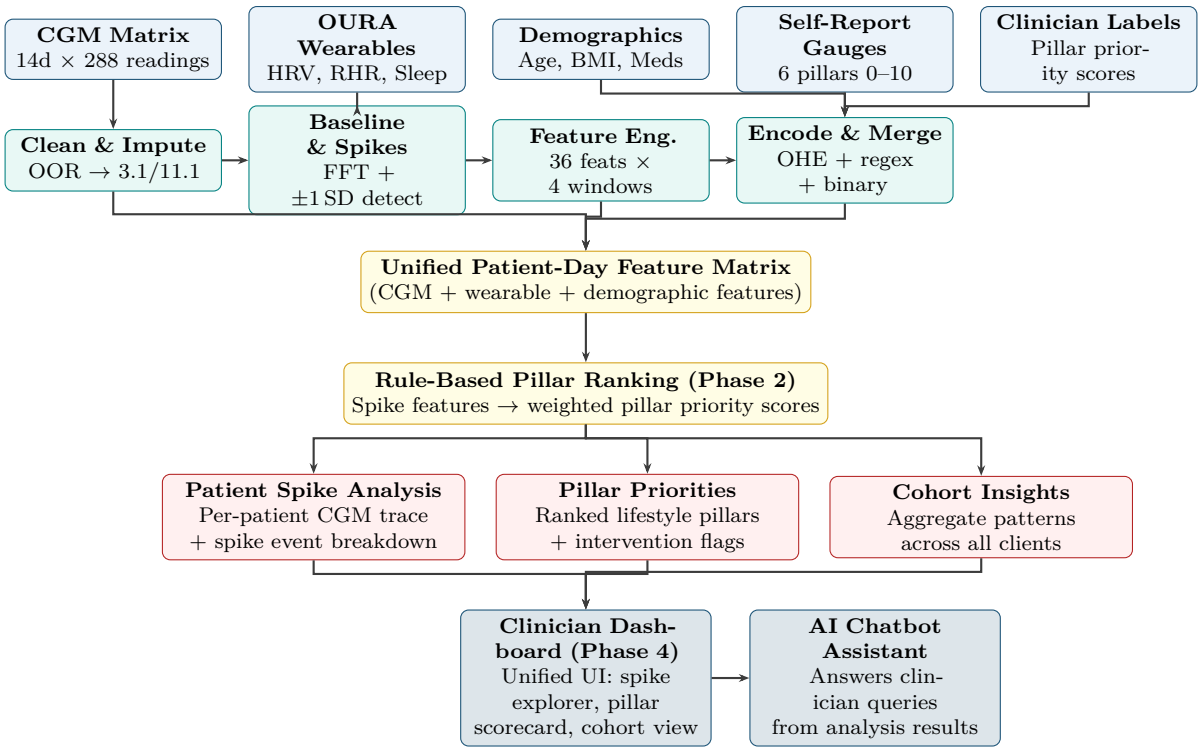
1. Problem Fit

Gillvray Health is an emerging lifestyle medicine providers from Cambridge. They collect 14-day Continuous Glucose Monitoring (CGM) data from their clients to prioritise lifestyle interventions across six pillars: ***sleep, nutrition, movement, stress, connection, and negative habits***. The challenge lies in extracting insights from these long and tedious CGM data rows. This bottleneck is compounded by the fact that existing CGM platforms collect only numerical metrics (TIR, HbA1c) and are blind to lifestyle context. A supervised ML approach grounded in personalised spike feature engineering is the correct tool: it learns relationships between intra-day glycaemic patterns and multi-label pillar priorities from clinician-labeled ground truth at a consistency and speed no manual review can match.

2. Market Comparison

Existing platforms (Levels, Dexcom Clarity, Abbott LibreView) deliver population-level dashboards with fixed thresholds. SPIKES differentiates via: (1) personalised per-patient SD-based spike detection; (2) multi-label lifestyle pillar prediction absent in all commercial tools; (3) a clinician-facing app prototype that distils spike features and pillar priority scores into an accessible, actionable UI — facilitating smoother clinician-patient interaction.

3. System Architecture



4. Technical Description

Data

The dataset comprises $n \approx 20$ clients: 14 days × 288 CGM readings (5-min, mmol/L); daily OURA wearables (HRV, RHR, sleep score, readiness); daily self-reported gauges across six pillars (0–10); clinician-reported pillar priority scores and binary intervention decisions.

Preprocessing

Raw CGM underwent: **(i)** physiological range filter (<2.0, >25.0 mmol/L removed); **(ii)** *Out of Range* imputation via last-valid-reading neighbourhood rule (3.1 if preceding value ≤ 4.0, else 11.1 mmol/L); **(iii)** short-gap linear interpolation (≤ 30 min). Demographics were one-hot encoded; medication data binarised (`is_on_Mounjaro`); health goals classified into 8 categories via regex.

Spike Feature Engineering

Personalised baselines follow Delannoy-Bruno et al. (2023): 30-minutes rolling median for intra-day spike detection; 24-hour rolling 40th-percentile (CGB) for daily fasting reference. Spike detection uses ±1 SD per patient-day (Service et al.,

1970; Fernandes et al., 2022), dual-tier (high-confidence & sensitive), bilateral (spikes & dips). Across four time windows (night 00–06h, morning 06–12h, afternoon 12–18h, evening 18–24h): mean/max amplitude; mean/max gradient (15-min rate-of-change (Rodbard, 2009)); mean/max duration; CV from baseline; iAUC (Guo et al., 2023); spike count — **36 CGM features** per patient-day.

Modelling

Features are merged into a unified patient-level matrix. A rule-based ranking system scores each of the six lifestyle pillars by aggregating weighted spike features — amplitude, gradient, duration, iAUC, and CV — across the four time windows. Pillar scores are normalised and ranked to produce a priority order for each patient. Given $n \approx 20$, the rule weights are validated against clinician-labelled priority scores using Spearman rank correlation and mean absolute rank error (MARE).

5. Integration & Roadmap

Current State

Data analysis is done by python in Jupyter notebooks, data extraction is done in python script, front-end of UI was done vite and tailwind.css with typescript. Inputs: `cgm_matrix.csv`, `patient_metadata.csv`, `patient_reported_metrics.csv`. Outputs: `SPIKES insight dashboard`, `AI chatbot assistant`, `pillar scores`.

Pathway

- **Near:** Package the Python analysis pipeline as a standalone executable; dashboard reads directly from pipeline output files with no external dependencies.
- **Medium:** Expand the UI to support multi-patient and multi-clinic deployments; add clinician annotation tools so feedback can be used to refine pillar weights over time.
- **Long:** Transition from rule-based ranking to a trained classifier as cohort size grows; federated learning across clinic sites; full NHS SNOMED-coded record integration.

Productionisation

- Containerise Python backend and Vite/TypeScript frontend as separate Docker services with pinned dependencies
- CI/CD via GitHub Actions; automated regression tests on both pipeline outputs and UI components
- Role-based access control for clinicians; GDPR-compliant audit log of all pillar scores and dashboard interactions
- Patient data encrypted at rest; regular penetration testing before clinical deployment

6. Trust, Privacy & Sustainability

Explainability

The rule-based ranking system is fully transparent by design — every pillar score traces directly back to specific spike features (e.g. elevated overnight iAUC drives Sleep priority, high post-meal amplitude drives Nutrition priority). The clinician dashboard displays the contributing features alongside each pillar score, so no prediction is ever a black box. This satisfies NHS and EU AI Act transparency requirements for clinical decision support.

Data Privacy

Patient identifiers are removed at ingestion (`DoB-REMOVED` in source data). All processing occurs locally — no patient data leaves the clinical environment. Synthetic IDs (`SPIKES-001..N`) replace originals throughout the pipeline. No third-party API calls are made with patient data.

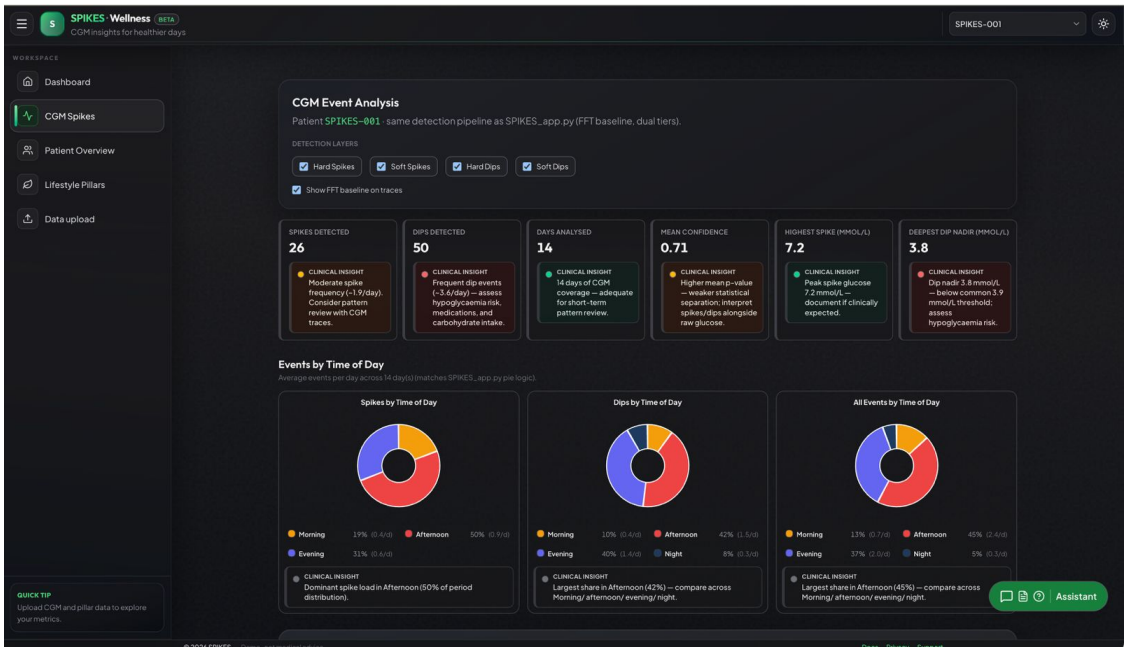
Cost & Time Saving

Manual MDT CGM review currently requires ∼30–45 min per patient. SPIKES generates a full pillar priority ranking and intervention shortlist in <3 seconds. At 20 clients per cohort this represents ∼**12 clinician-hours saved per cycle**, redirecting MDT capacity from data interpretation to direct patient care.

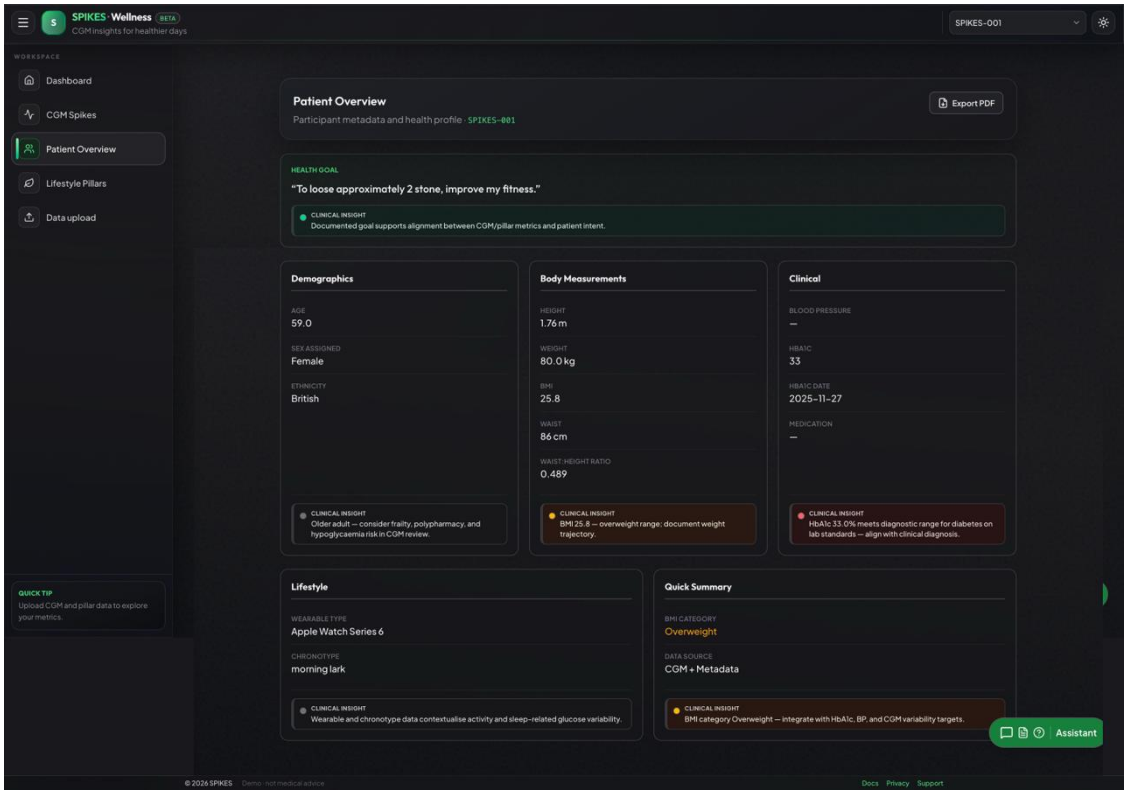
Sustainability

The pipeline runs entirely on commodity hardware with no GPU requirement. The rule-based core has negligible compute cost compared to deep learning alternatives. As the patient registry grows, the weighting system can be recalibrated without full retraining, keeping operational costs flat.

7. Demo Images



(1) CGM Event Analysis — Per-patient spike and dip detection with FFT baseline, confidence scores, and time-of-day distribution charts.



(2) Patient Overview — Demographics, body measurements, clinical markers, wearable type, chronotype, and automated clinical insights per patient.



(3) SPIKES AI Assistant — Clinician chatbot aware of the loaded session data; answers natural language queries about spike events, pillar scores, and cross-patient patterns.

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